

1. A particle is moving in x - z plane such that its x and z coordinate varies with time t as $x = (3t + 2t^2 + 1)$ m and $z = (8t^2)$ m (where time t is in second). The ratio of magnitude of acceleration of particle in x and z direction at any time t is:

- (A) $\frac{1}{3}$
(B) $\frac{2}{3}$
(C) $\frac{1}{4}$
(D) $\frac{3}{4}$

2. If rain is falling along $-y$ axis with a velocity $\vec{V}_R$ and an observer is moving along $-x$ axis with speed $\vec{V}_M$, then the velocity of rain relative to observer in vector notation will be;

- (A) $\vec{V}_{RM} = V_R \hat{j} - V_M \hat{i}$
(B) $\vec{V}_{RM} = -V_R \hat{j} + V_M \hat{i}$
(C) $\vec{V}_{RM} = V_R \hat{j} + V_M \hat{i}$
(D) $\vec{V}_{RM} = -V_R \hat{j} - V_M \hat{i}$

3. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The maximum horizontal range of projectile is proportional to square of velocity.
Reason R: The maximum horizontal range of projectile is equal to maximum height attained by projectile.

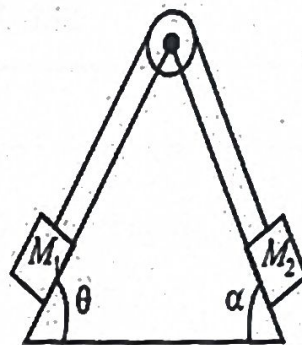
In the light of the above statements, choose the correct answer from the options given below:

- (A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true and R is NOT the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

4. A boy stands on a weighing machine which is placed on a table and ties it with his feet. The reading on the machine shows 200 N. Now he attempts to jump from a table on ground with the machine tied. The reading on the machine while he is in air will be: ($g = 10 \text{ m/s}^2$)

- (A) 200 N (B) 20 N
(C) 2000 N (D) zero

5. Two masses M_1 and M_2 connected by means of a string which is made to pass over light, smooth pulley are in equilibrium on a fixed smooth wedge as shown in figure. If $\theta = 60^\circ$ and $\alpha = 30^\circ$, then the ratio of M_1 to M_2 is



- (A) 1:2 (B) $2:\sqrt{3}$
(C) $1:\sqrt{3}$ (D) $\sqrt{3}:1$

6. A car starting from rest accelerates at the rate f through a distance s , then continues at constant speed for time t and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is $15s$, then $s = ?$

- (A) $s = \frac{ft^2}{72}$
(B) $s = \frac{ft^2}{4}$
(C) $s = \frac{ft^2}{6}$
(D) $s = \frac{ft^2}{2}$

7. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: An object can have constant speed but variable velocity.

Reason R: Speed is a scalar but velocity is a vector quantity.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true and R is NOT the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.



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A particle located at origin at time $t = 0$, starts moving along positive y direction with a velocity v that varies as $v = \alpha\sqrt{y}$. The displacement of the particle varies with time as; (α is constant)

- (A) $\frac{t^3}{3}$ (B) $\frac{t^{-2}}{2}$
(C) $\frac{t^2}{4}$ (D) $2t$

9. A particle of unit mass undergoes one dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$ where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x , is given by:

- (A) $-2n\beta^2 x^{-4n-1}$ (B) $-2\beta^2 x^{-2n+1}$
(C) $-2n\beta^2 x^{-4n+1}$ (D) $-2n\beta^2 x^{-2n-1}$

10. Given below are two statements:

Statement I: Kinetic friction is a self-adjusting force.

Statement II: Friction does not depend upon mass of the body.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Statement I is correct but Statement II is incorrect.
(B) Statement I is incorrect but Statement II is correct.
(C) Both Statement I and Statement II are correct.
(D) Both Statement I and Statement II are incorrect.

11. A block of mass m is released from rest at a height R above a horizontal surface. The acceleration due to gravity is g . The block slides along the inside of a frictionless circular hoop of radius R . What is the magnitude of the normal force exerted on the block by the hoop when the block has reached the bottom?

- (A) mg
(B) $3mg$
(C) $\frac{mg^2}{R}$
(D) $2mg$

12. Power applied to a particle varies with time as $P = (4t^3 - 5t + 2)$ watt, where t is in second. Find the change in its K.E. between time $t = 2$ and $t = 4$ sec.

- (A) 212 J (B) 213 J
(C) 214 J (D) 215 J

13. Repeated observation in a experiment gave the values 2.29, 2.24 and 2.34. The mean absolute error in this experiment.

- (A) $\frac{1}{10}$ (B) $\frac{1}{20}$
(C) $\frac{1}{30}$ (D) $\frac{1}{40}$

14. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of -0.004 cm, the correct diameter of the ball is

- (A) 0.521 cm (B) 0.525 cm
(C) 0.053 cm (D) 0.529 cm

15. Considering force (F), Velocity (V) and energy (E) as fundamental quantities, match the correct dimensions of following quantities.

List-I		List-II	
(A)	Mass	(I)	$[F^{-1}V^0E^1]$
(B)	Light year	(II)	$[F^1V^1E^{-1}]$
(C)	Frequency	(III)	$[F^3V^0E^{-2}]$
(D)	Pressure	(IV)	$[F^0V^{-2}E^1]$

Choose the correct answer from the options given below:

- (A) A-IV, B-I, C-II, D-III
(B) A-I, B-III, C-II, D-IV
(C) A-II, B-I, C-III, D-IV
(D) A-IV, B-I, C-III, D-II

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16. Rutherford's experiment on scattering of α -particles showed for the first time that the atom has:
- (A) Electrons (B) Protons
(C) Neutrons (D) Nucleus

17. Among the following, the molecule that will have the highest dipole moment is:
- (A) H_2 (B) HI
(C) HBr (D) HF

18. Match List-I with List-II.

List-I (Element)		List-II (Group number)	
(A)	N	(I)	2
(B)	Be	(II)	3
(C)	Sc	(III)	17
(D)	Cl	(IV)	15

Choose the correct answer from the options given below:

- (A) A-I, B-II, C-IV, D-III
(B) A-II, B-I, C-III, D-IV
(C) A-II, B-I, C-IV, D-III
(D) A-IV, B-I, C-II, D-III

19. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The number of oxygen atoms in 1 mole of H_2SO_4 is $4 \times$ Avogadro's number.
Reason R: Each molecule of H_2SO_4 contains 4 oxygen atoms.

In the light of the above statements, choose the correct answer from the options given below:

- (A) A is true but R is false.
(B) A is false but R is true.
(C) Both A and R are true and R is the correct explanation of A.
(D) Both A and R are true but R is not the correct explanation of A.

20. In magnesium atom, energy of an electron in an orbital is determined by quantum number(s):
- (A) $n + l$ (B) l only
(C) m (D) s

21. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): CO_2 contains polar covalent bonds.
Reason (R): CO_2 is a linear molecule.
In the light of the above statements, choose the correct answer from the options given below:

- (A) A is true but R is false.
(B) A is false but R is true.
(C) Both A and R are true and R is the correct explanation of A.
(D) Both A and R are true but R is NOT the correct explanation of A.

22. Neutral oxide among the following is;
- (A) NO (B) SO_2
(C) N_2O_3 (D) SO_3

23. A compound made of two elements A and B are found to contain 25% A (atomic mass 12.5) and 75% B (atomic mass 37.5). The simplest formula of the compound is;
- (A) AB (B) AB_2
(C) AB_3 (D) A_3B

24. If the wavelength of green light is about $5000\text{\AA}$, then the frequency of its wave is:
- (A) $16 \times 10^{14} \text{ sec}^{-1}$ (B) $16 \times 10^{-14} \text{ sec}^{-1}$
(C) $6 \times 10^{14} \text{ sec}^{-1}$ (D) $6 \times 10^{-14} \text{ sec}^{-1}$

25. To which of the following species is the octet rule applicable?
- (A) BrF_3 (B) SF_6
(C) IF_7 (D) CO_2

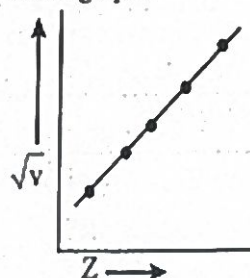
26. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Bond energy order:
 $C-C < C=C < C \equiv C$.
Reason R: Bond energy increases with increase in bond order.

In the light of the above statements, choose the correct answer from the options given below:

- (A) A is true but R is false.
(B) A is false but R is true.
(C) Both A and R are true and R is the correct explanation of A.
(D) Both A and R are true but R is NOT the correct explanation of A.

27. When the azimuthal quantum number has the value of 2, the number of orbitals possible are:
- (A) 3 (B) 0
(C) 7 (D) 5

28. According to Henry Moseley, what does Z represent in the graph?



- (A) Atomic mass (B) Atomic number
(C) Mass number (D) Number of neutrons

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Consider the following statements regarding hybridisation:

1. In sp hybridisation, two hybrid orbitals are formed.
2. In sp^2 hybridisation, geometry is trigonal planar.
3. In sp^3 hybridisation, bond angle is always 109.5° .
4. Hybrid orbitals are equivalent in energy and shape.

Choose the correct option:

- (A) 2 and 3 only
(B) 1, 2, 3 and 4 are correct
(C) 1, 2 and 4 only
(D) 1 and 4 only

30. Consider the following statements:

1. Principal quantum number determines the size of the orbital.
2. Azimuthal quantum number determines the shape of the orbital.
3. Magnetic quantum number determines the orientation of the orbital.
4. Spin quantum number can have values 0 and 1.

Choose the correct option:

- (A) 1, 2 and 3 are correct
(B) 1 and 4 are correct
(C) 2, 3 and 4 are correct
(D) All are correct

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31. Identify the incorrect statement about nucleolus.
 (A) It is a spherical structure present in the nucleoplasm.
 (B) It is a membrane bound structure.
 (C) It is a site for active ribosomal RNA synthesis.
 (D) Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.
32. Unicellular organisms:
 (A) are capable of independent existence.
 (B) are dependent on others for survival.
 (C) are unable to perform essential functions of life.
 (D) are composed of many cells.
33. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Centromere holds four chromatids of a chromosome.
Reason R: In acrocentric chromosomes, the centromere is situated close to its end forming one extremely short and one very long arm.
 In the light of the above statements, choose the correct answer from the options given below:
 (A) A is true but R is false.
 (B) A is false but R is true.
 (C) Both A and R are true and R is the correct explanation of A.
 (D) Both A and R are true but R is NOT the correct explanation of A.
34. Identify from the following that is/are found in algal cell walls.
 I. Hemicellulose
 II. Cholesterol
 III. Galactans
 IV. Calcium carbonate
 V. Proteins
 Choose the most appropriate answer from the options given below:
 (A) III and IV only
 (B) All except V
 (C) II, III and IV only
 (D) I and III only
35. Binomial system of nomenclature means that every organism has:
 (A) two names, one scientific and one popular.
 (B) one name given by two scientists.
 (C) one scientific name consisting of a generic and a specific epithet.
 (D) two names, one denoting the latinised name of the place and the other of the person, who described it.
36. Choose odd one out w.r.t. *Panthera leo*.
 (A) Common name of tiger
 (B) *Panthera* represents generic name
 (C) *leo* represents specific epithet
 (D) *Panthera* represents higher level of taxon than *leo*
37. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Meiosis involves two sequential cycles of nuclear and cell division called meiosis-I and meiosis-II.
Reason R: Meiosis involves pairing of homologous chromosomes and recombination between non homologous chromosomes.
 In the light of the above statements, choose the correct answer from the options given below.
 (A) A is true but R is false.
 (B) A is false but R is true.
 (C) Both A and R are true and R is the correct explanation of A.
 (D) Both A and R are true but R is NOT the correct explanation of A.
38. In anaphase of mitosis,
 (A) chromosomes get arranged in middle of cell.
 (B) nuclear envelope disappears.
 (C) chromosome fibres become clear.
 (D) chromatids move to opposite poles.
39. Select the correct statements with respect to the cell cycle.
 I. The interphase lasts not more than 50% of the duration of cell cycle.
 II. The plants can show mitotic divisions in both haploid and diploid cells.
 III. G₂ phase corresponds to the interval between mitosis and initiation of DNA replication.
 IV. S or synthesis phase marks the period during which DNA synthesis or replication takes place.
 Choose the most appropriate answer from the options given below:
 (A) I and III only (B) I and IV only
 (C) I and II only (D) II and IV only
40. All are diseases caused by prions, except:
 (A) bovine spongiform encephalopathy.
 (B) mad cow disease.
 (C) Creutzfeldt-Jacob disease.
 (D) influenza.

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Which of the following is an incorrect characteristic of bacteria?

- (A) They cannot live in extreme habitats such as hot springs, deserts, snow and deep oceans.
- (B) They show most extensive metabolic diversity.
- (C) Most of them are heterotrophic.
- (D) They reproduce mainly by fission.

42. The parasitic fungus on mustard plant causing white rust belongs to which class of fungi?

- (A) Phycomycetes
- (B) Ascomycetes
- (C) Basidiomycetes
- (D) Deuteromycetes

43. Select the correct statements.

- A. Bladderwort and Venus fly trap are examples of insectivorous plants.
- B. *Cuscuta* is a parasite.
- C. Kingdom plantae includes algae, bryophytes, pteridophytes, gymnosperms and angiosperms.
- D. Kingdom animalia includes chlorophyll-containing organisms.

Choose the most appropriate answer from the options given below:

- (A) A and B only
- (B) B and C only
- (C) C and D only
- (D) A, B and C only

44. Under favourable conditions, slime moulds form:

- (A) plasmodium.
- (B) conidia.
- (C) fruiting bodies.
- (D) zygote.

45. Which of the following statement is incorrect regarding Chrysophytes?

- (A) They include both diatoms and desmids.
- (B) They are microscopic and float passively in water currents.
- (C) They belong to Kingdom Protista.
- (D) Most of them are heterotrophs.

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46. Which of the following is not supported by cartilaginous rings?

- (A) Primary bronchi
- (B) Tertiary bronchi
- (C) Trachea
- (D) Alveoli

47. What is the typical range of leucocytes per cubic millimeter in human blood?

- (A) 1,500 - 3,500
- (B) 4,000 - 11,000
- (C) 6,000 - 8,000
- (D) 10,000 - 15,000

48. Which of the following acids is used to analyze the chemical composition of living tissues?

- (A) HCl
- (B) Cl_3CCOOH
- (C) Acetic acid
- (D) H_2SO_4

49. Blood leaving the tissues and going towards the heart (systemic veins) has:

- (A) $\text{pO}_2 = 95 \text{ mm Hg}$, $\text{pCO}_2 = 40 \text{ mm Hg}$
- (B) $\text{pO}_2 = 40 \text{ mm Hg}$, $\text{pCO}_2 = 45 \text{ mm Hg}$
- (C) $\text{pO}_2 = 104 \text{ mm Hg}$, $\text{pCO}_2 = 40 \text{ mm Hg}$
- (D) $\text{pO}_2 = 40 \text{ mm Hg}$, $\text{pCO}_2 = 40 \text{ mm Hg}$

50. Each nucleotide is made up of:

- (A) Nitrogen containing carbohydrate and pentose sugar with phosphate.
- (B) Cyclic nitrogenous base + pentose sugar and a phosphate group.
- (C) Nitrogenous base + hexose sugar + phosphate.
- (D) Amino sugar + ribose + phosphate

51. Which of the following best describes pulmonary circulation?

- (A) It carries oxygenated blood from the left ventricle to the aorta.
- (B) It involves the movement of deoxygenated blood from the right ventricle to the lungs.
- (C) It collects nutrients and oxygen from the tissues.
- (D) It returns deoxygenated blood to the left atrium.

52. Which of the following constitutes the exchange part of the human respiratory system?

- (A) External nostrils to trachea
- (B) External nostrils to primary bronchioles
- (C) Alveoli and their ducts
- (D) External nostrils to terminal bronchioles

53. How do parasympathetic neural signals affect the working of the heart?

- (A) Reduce both heart rate and cardiac output.
- (B) Heart rate is increased without affecting the cardiac output.
- (C) Both heart rate and cardiac output increase.
- (D) Heart rate decreases but cardiac output increases.

54. Which of the following is the correct statement for respiration in human beings?

- (A) About 90 percent carbon dioxide is carried by haemoglobin as carbamino haemoglobin.
- (B) Neural signals from optic lobe of brain can increase the duration of inspiration.
- (C) Workers in grinding and stone-breaking industries may suffer, from lung fibrosis.
- (D) All of these

55. Match List-I with List-II:

List I		List II	
(A)	Apoenzyme	(I)	Forms coordination bond with the side chain at the active site.
(B)	Prosthetic group	(II)	Transiently associated with the apoenzyme.
(C)	Co-enzyme	(III)	Protein part of the enzyme.
(D)	Metal ion	(IV)	Tightly bound to the apoenzyme.

Choose the correct answer from the options given below.

- (A) A-I, B-II, C-III, D-IV
- (B) A-IV, B-II, C-I, D-III
- (C) A-III, B-IV, C-II, D-I
- (D) A-III, B-I, C-II, D-IV

56. How many of the following statements are correct?

- I. About 98% of mass of every living organism is composed of just six elements including C, H, O, N and phosphorus and sulphur.
- II. Chitin is a homopolymer of N-acetyl glucosamine.
- III. Glycine is a sulphur containing amino acid.
- IV. Arachidonic acid has 20 carbon atoms including the carboxyl carbon.
- V. Oils have a lower melting point.
- VI. The neural tissues have lipids with more complex structures.

- (A) Three
- (B) Four
- (C) Five
- (D) Six

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Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: A healthy individual has 12-16g of haemoglobin in every 1000 ml of blood.

Reason R: Haemoglobin molecules play a significant role in transport of respiratory gases like oxygen.

In the light of the above statements, choose the correct answer from the options given below:

- (A) A is true but R is false.
- (B) A is false but R is true.
- (C) Both A and R are true and R is the correct explanation of A.
- (D) Both A and R are true but R is not the correct explanation of A.

58. Read the following statements (I-IV).

- I. Plasma is a straw coloured, viscous fluid constituting nearly 45 percent of the blood.
- II. Leucocytes play a significant role in transport of respiratory gases.
- III. Serum is the plasma without the clotting factors.
- IV. Thrombocytes are cell fragments produced from megakaryocytes.

Which of the following statements are incorrect?

- (A) II and III only
- (B) III and IV only
- (C) I and IV only
- (D) I and II only

59. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Nucleic acids are present in the acid insoluble fraction of living tissue.

Reason R: Nucleic acids are micromolecules.

In the light of the above statements, choose the correct answer from the options given below:

- (A) A is true but R is false.
- (B) A is false but R is true.
- (C) Both A and R are true and R is the correct explanation of A.
- (D) Both A and R are true but R is not the correct explanation of A.

60. Which of the following factors are favourable for the formation of oxyhaemoglobin?

- I. High pO_2 , low pCO_2
- II. Lesser H^+ concentration, lower temperature
- III. Low pO_2 , high pCO_2
- IV. High H^+ , higher temperature

- (A) Only I
- (B) I and II
- (C) II and III
- (D) III and IV

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